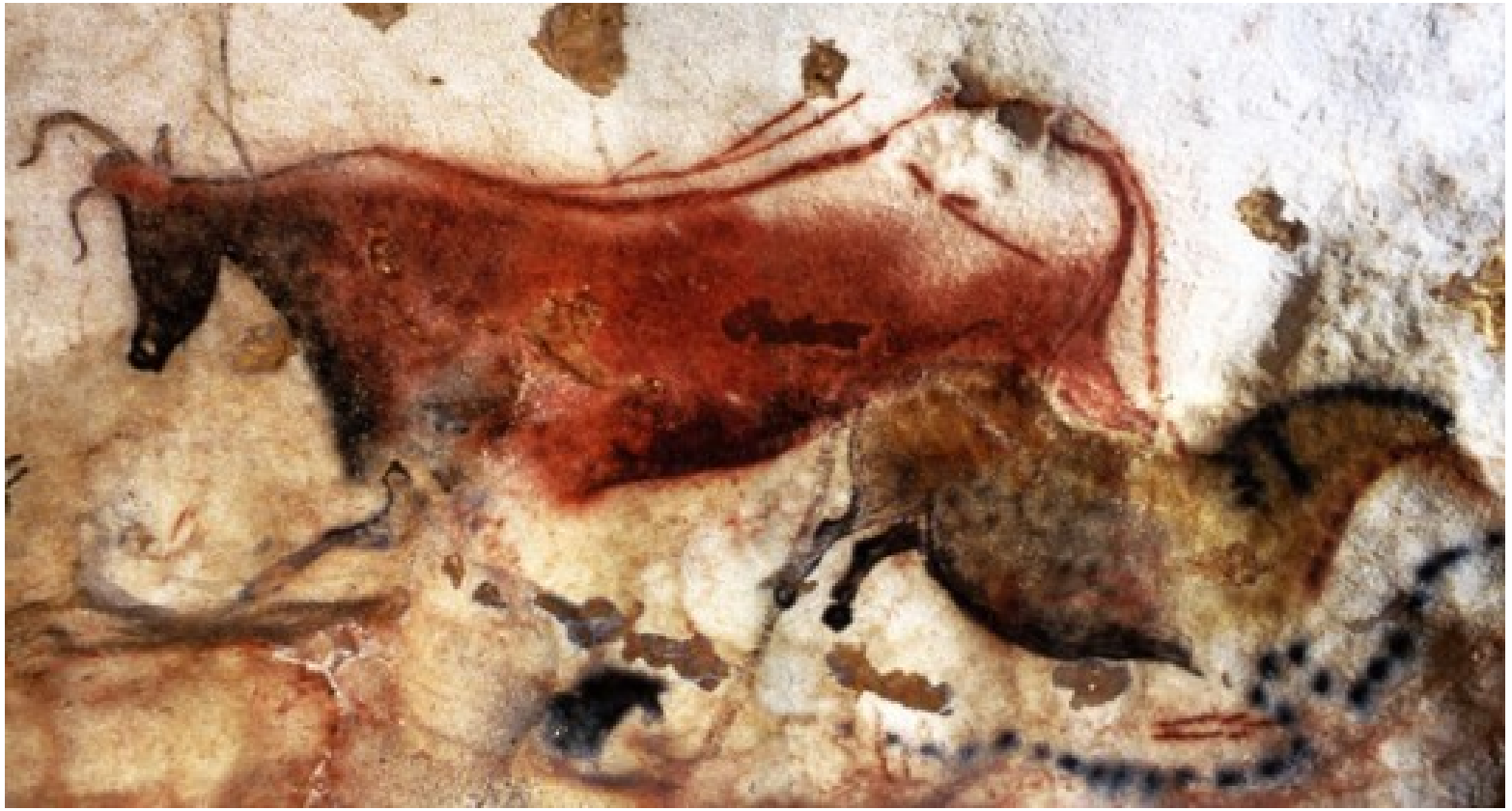
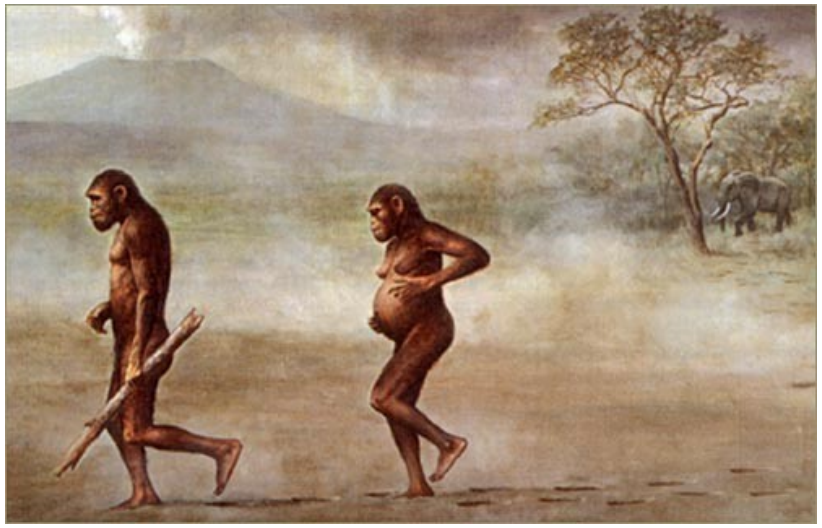
Hominin Cultural Evolution



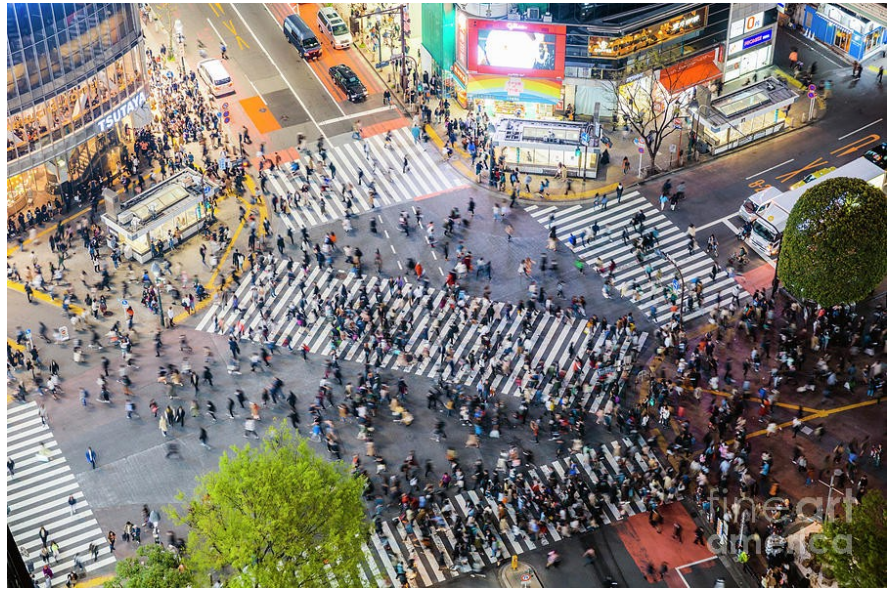
Cultural Evolution

Introduction



- In addition to anatomical and physiological evolution, hominins have developed complex cultures that allow them to survive better in their environment.

- Bipedalism came first, followed by brain size, which in turn led to innovation and adaptation through abstract thought and problem solving.



- Cultural evolution can be seen in:
  - Tool and technology development
  - Development of methods for obtaining food
  - Increasing sophistication of language
  - Other changes including symbolic representation (eg visual art)

Cultural Evolution – *Australopithecus*

- Brain size and evidence of diet show likely to have culture and lifestyle similar to chimps but adapted to grassland.
  - Had home bases: lived in one place, left from there to forage for food
  - No evidence that they built shelters
  - No evidence of fire use or cooking
  - No evidence of language – likely similar communication to chimps
  - No evidence of art
  - No evidence of abstract or symbolic thought
  - No evidence of hunting; primarily foraged, plant-based diet.



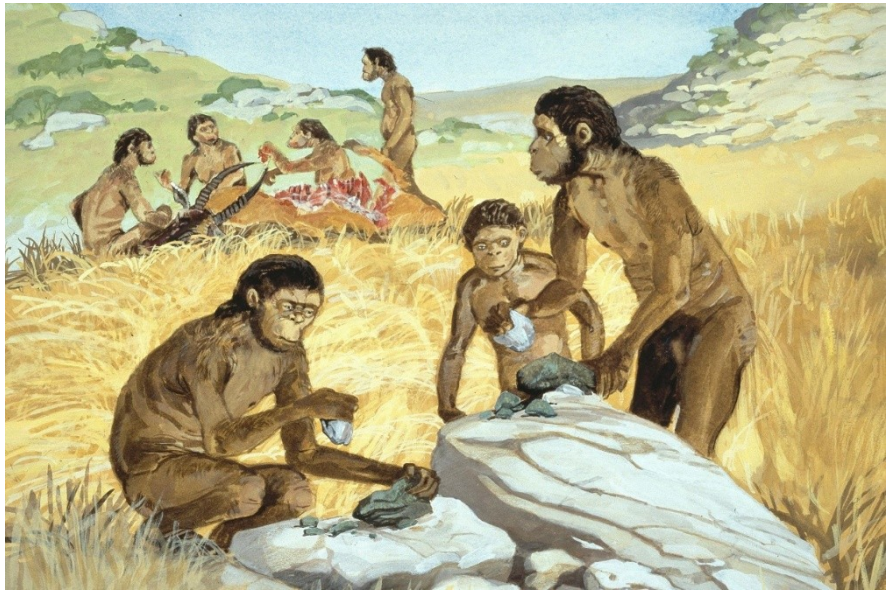
- Tools
  - Used rocks and sticks as tools, for example to break hard-shelled fruits and seeds.
  - **May** have made primitive stone tools (eg Oldowan tools) but experts are divided on this – not enough evidence. *In a short answer question asking about use of Oldowan tools, say that Australopithecus may have used them, and H. habilis definitely did.*

Cultural Evolution - *Homo habilis*

- Larger brains than Australopithecines
  - Needed more complex fats and higher levels of protein – brains use energy!
  - Therefore needed to consume meat
  - May have had some simple language development – not a lot of evidence

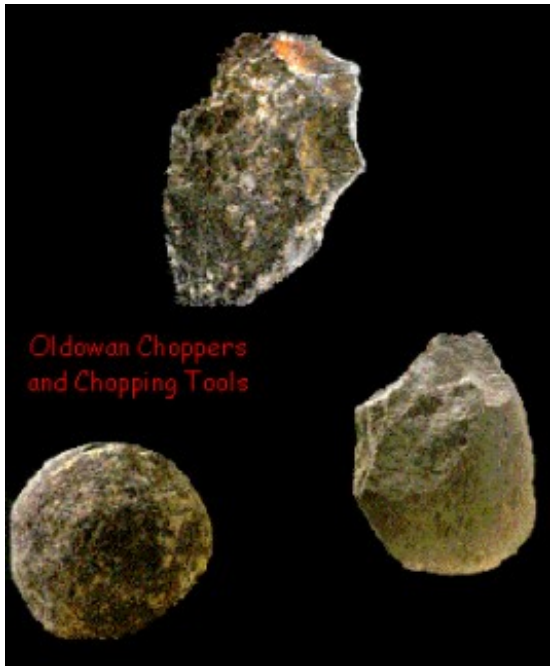
- Home bases – left to forage for plant food and scavenge animal remains
  - No evidence of built shelters
  - No evidence of fire use
  - No evidence of art
  - No evidence of abstract or symbolic thought

- Tool Use – Oldowan Tool Culture
  - Made and used this tool type
  - Range of pebble tools found: choppers, scrapers, flakers, chisels
  - Require precision grip
  - Used tools to extract marrow from animal bones
  - Tool use enabled colonisation of a broader range of habitats
  - May have been the first to make these tools (some experts say *Australopithecus* were first to make them.)

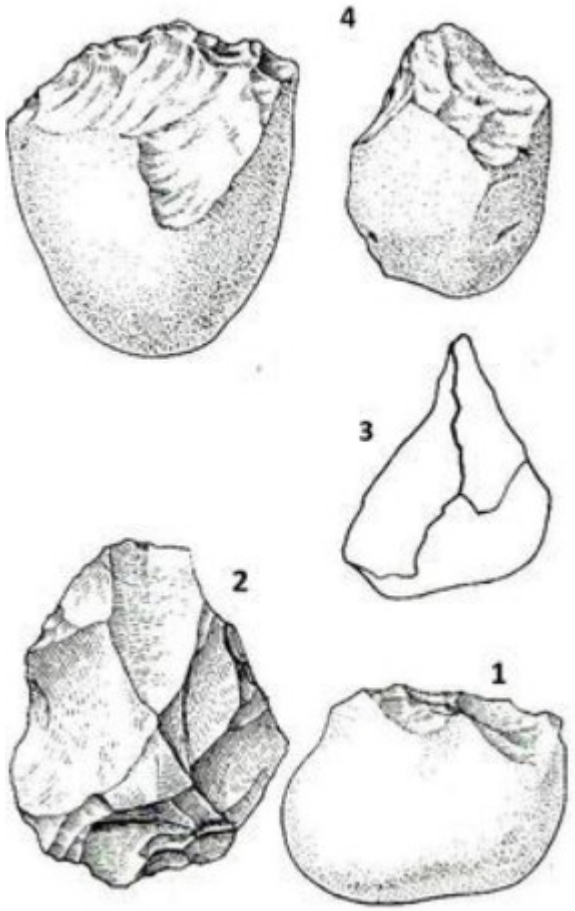




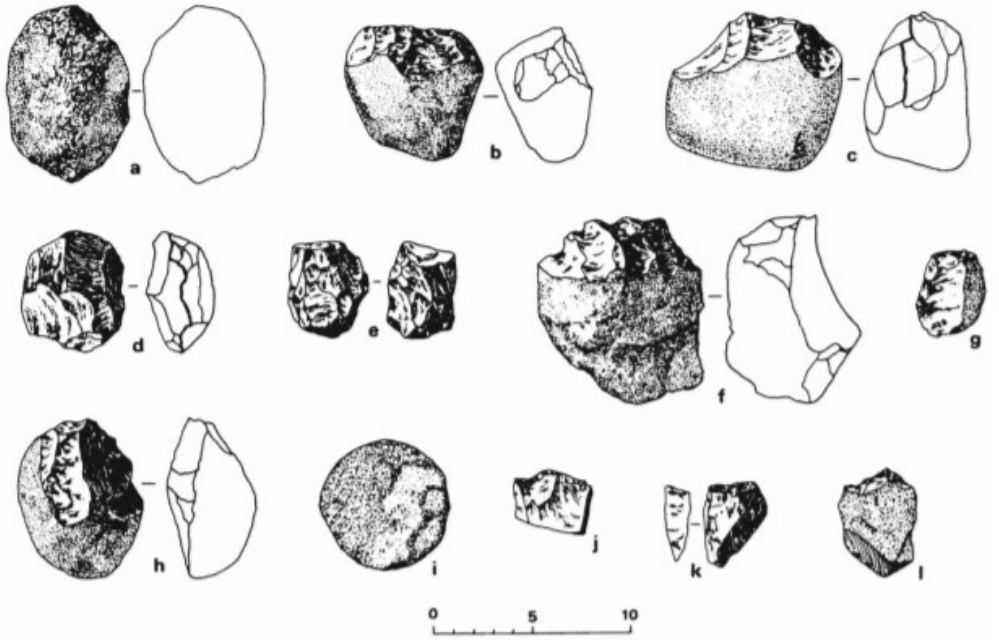
Cultural Evolution - *Homo habilis* – Features of Oldowan tools







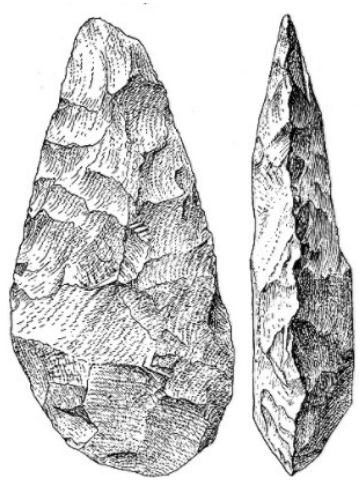
- Large tools – size of hand
- Rough flaking, chips bashed off to make sharp edge.
- Generalised tools for hitting things until they break. Not developed for specific purpose (no arrowheads, fish hooks, blades etc)



Cultural Evolution - *Homo erectus*

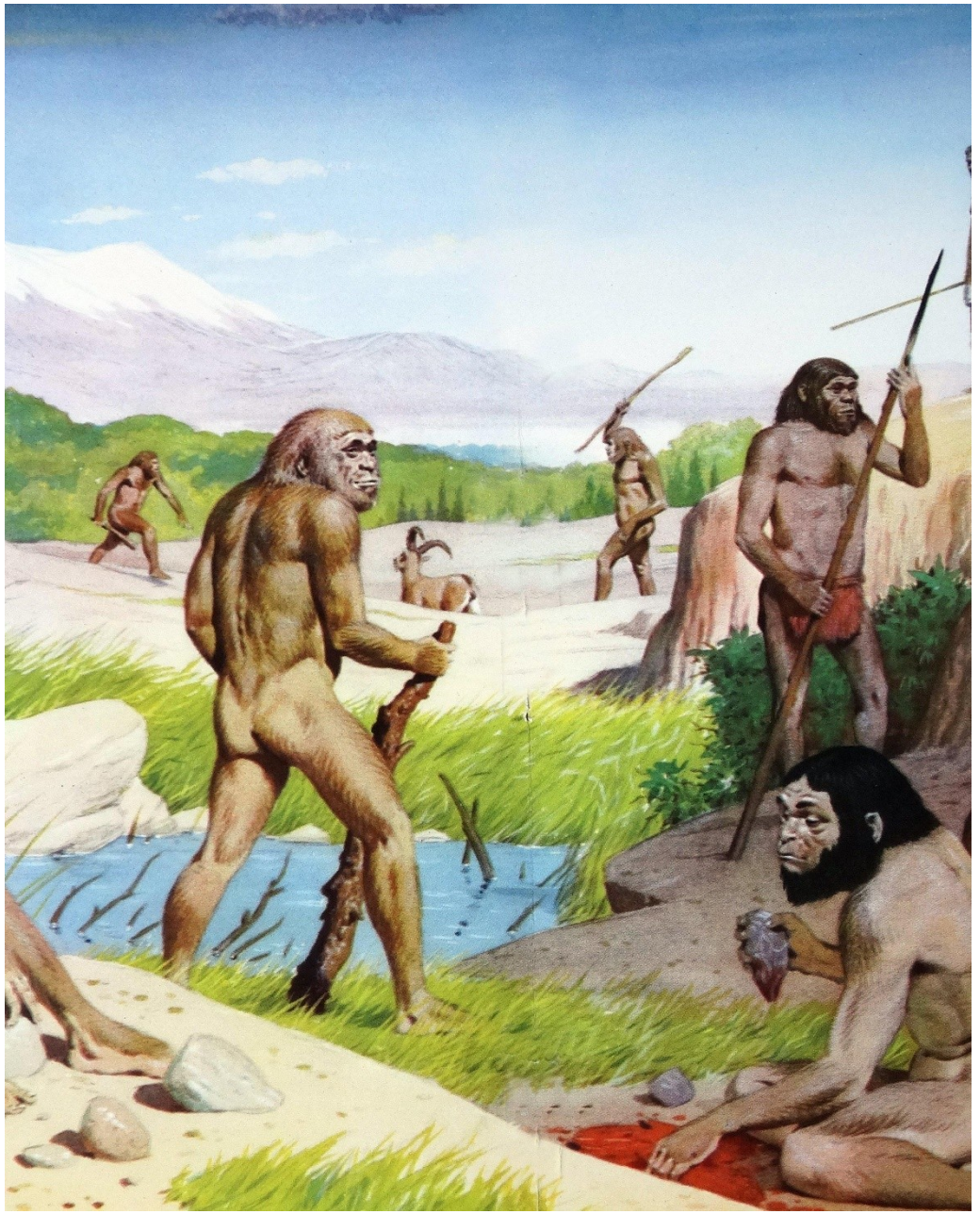


- Further increase in brain size and development
- Began to modify environment to suit own purposes
- Home bases:
  - Built shelters to protect from the elements
  - *Terra Amata*: site dating to 400 000 years ago
    - Wood huts
    - Evidence of fire use
    - Tools from stone
    - Remnants of animal bone from hunting - also fish bones – fishing?
- Evidence that *H.erectus* used fire:
    - Kept predators away
    - Warmth and light
    - Used during hunting to drive animals
    - Cooking of food
- Tools
  - Flaked around all edges, teardrop shaped “hand axes”
  - Referred to as *Acheulean tools* (from where they were first discovered)
  - Only appeared to make this one type of tool, and possibly wooden spears

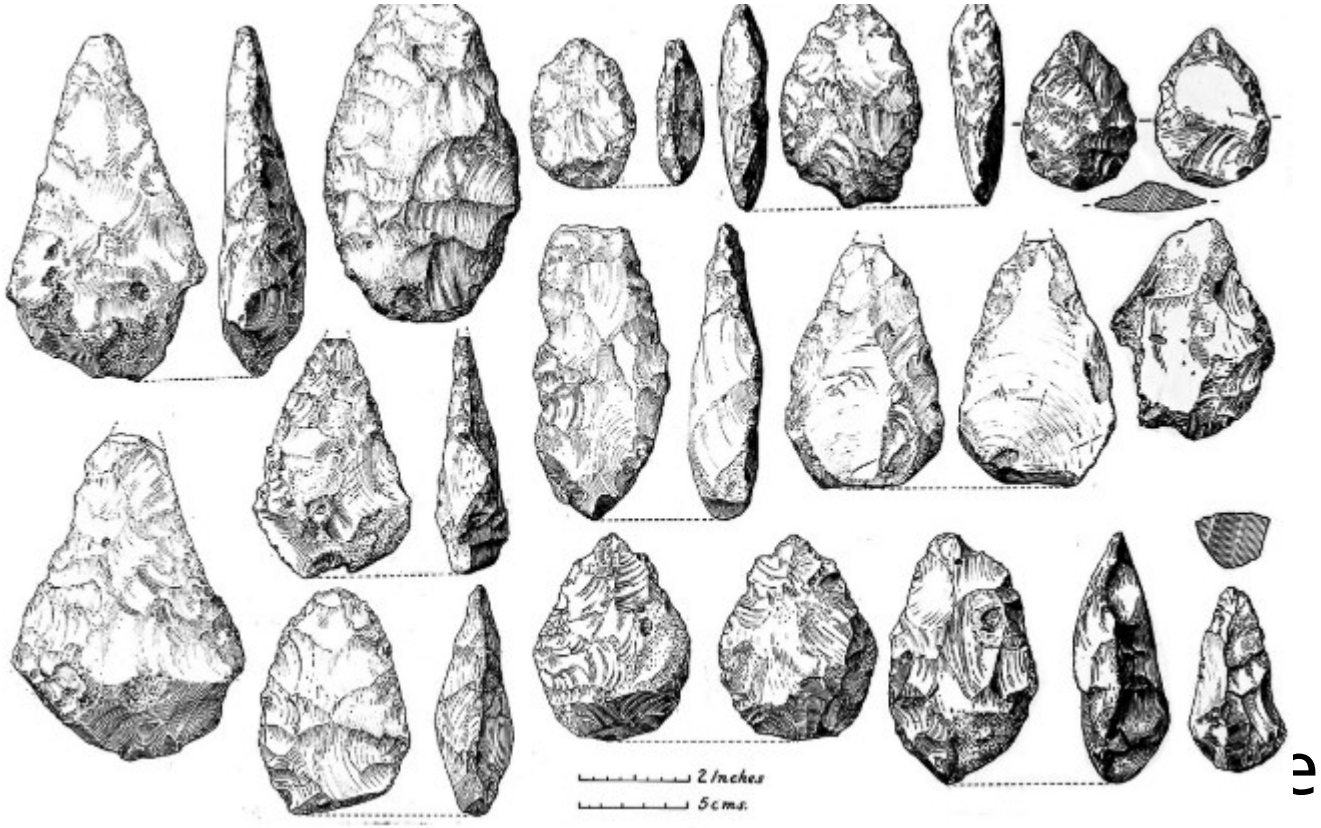


Cultural Evolution – *Homo erectus*

- Food:
  - Hunter- gatherer society – collaboration and cooperation
  - Most food from gathering, supplemented by hunting
  - Skilful, strategic hunters
    - Evidence of planning – planned baboon slaughter miles away from base
    - Drove elephants into swamps using fire, then butchered them
    - Sophisticated, systematic butchering using tools
  - Women used complex techniques and collaboration to gather food.
    - Balanced with child rearing
- Social organisation
  - A result of hunting/gathering/use of tools and fire.
  - Complex society, shared care, food sharing.
  - Relatively complex social language may have developed.



Cultural Evolution - *Homo erectus* – Features of Acheulean tools



- Large – hand sized
- Rough tear-drop shape
- Not a lot of variation
- General rather than specific use
- Finer flaking than Oldowan
- Bifacial – thin axe-like edges with two broader faces.

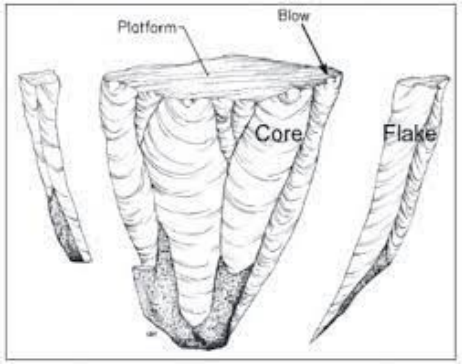
ectus

Cultural Evolution: *Homo neanderthalensis*

- Specialised hunters
  - Meat an important part of diet
  - Plant food sources scarce in ice-bound Europe
  - Needed energy from animal fats to survive the cold
- Made sophisticated stone tools
  - “Mousterian Industry”
  - Flaked a range of tool types from stone cores – many tools from a single core
  - Hafted (stuck) stone tools to wood to make axes, spears.
  - Made stone scrapers to scrape hides, then pierced them with sharp tools and bound them together with strips of hide to make clothing.
  - Previous hominins made simple, generalised tools and did not use the core and flaking technique to make many different tools from a single stone
- Made clothing
- Likely to have lived in caves
- Could make and use fire for hunting, cooking and warmth
- Buried dead, cared for sick and injured
- No evidence of abstract thought or symbolic representation eg cave art.





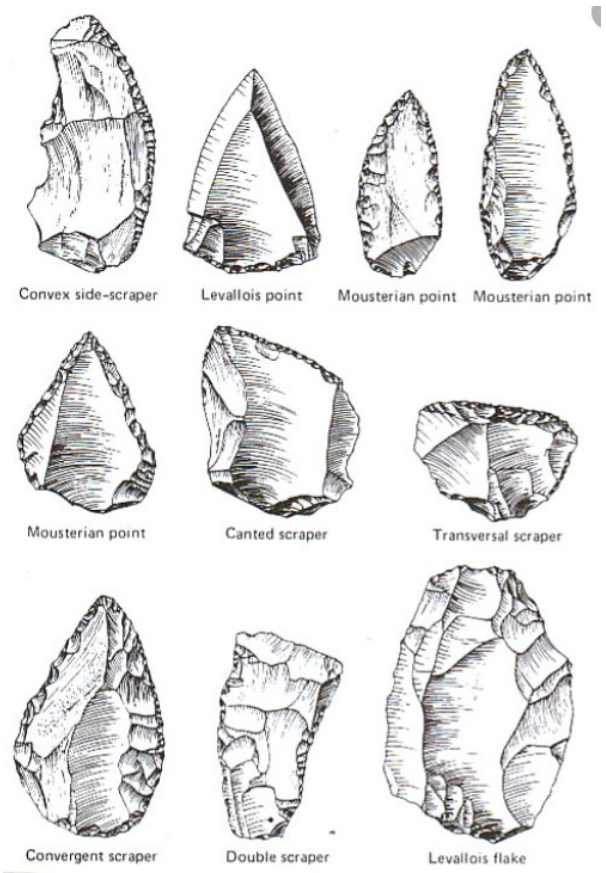


Cultural Evolution – *Homo neanderthalensis* – Tools



Figure 20.7 Neanderthal tools

- Variety of tool shapes and sizes to suit different purposes eg scraping, stabbing, piercing
- Relatively fine flaking to edges for sharpness
- Points would have been hafted to wood for use as spears



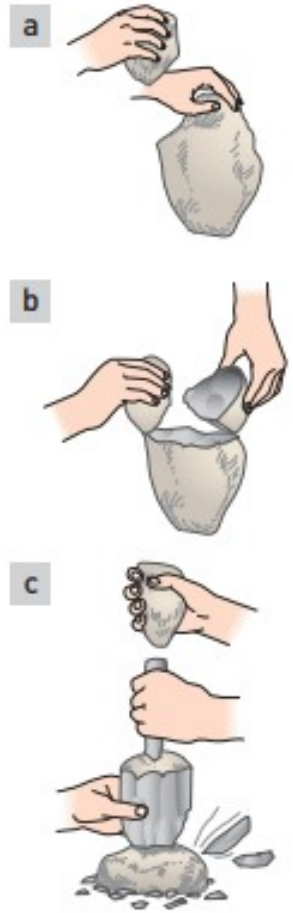
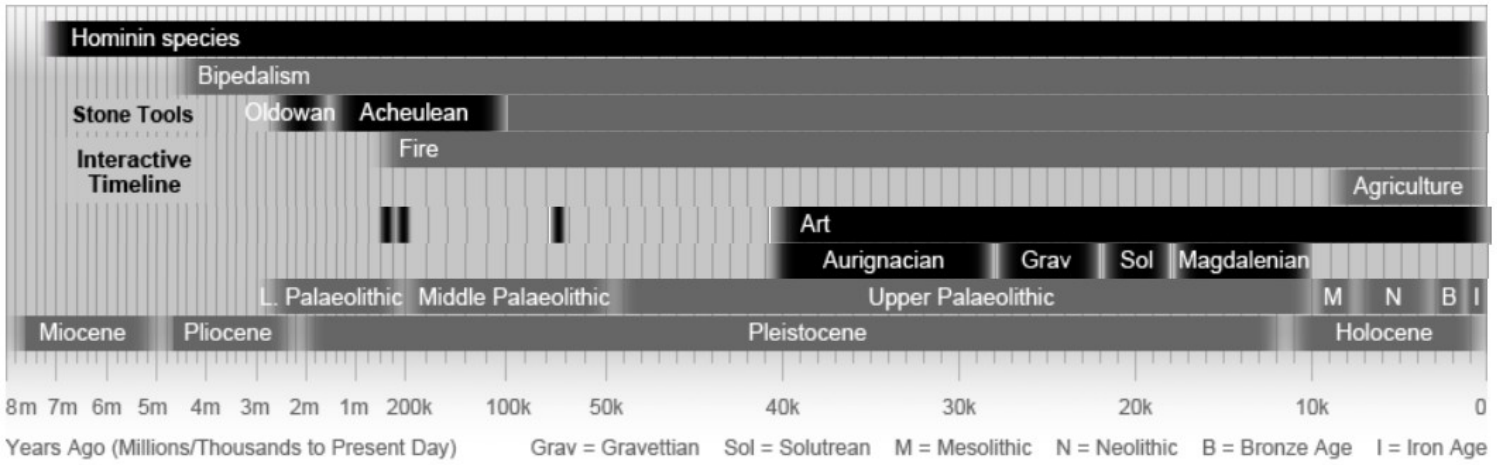


Figure 20.6 The production of flake tools: **a** and **b** show preparation of the core, and **c** shows how a large number of flakes can be produced.

Cultural Evolution – *Homo sapiens* – middle Palaeolithic

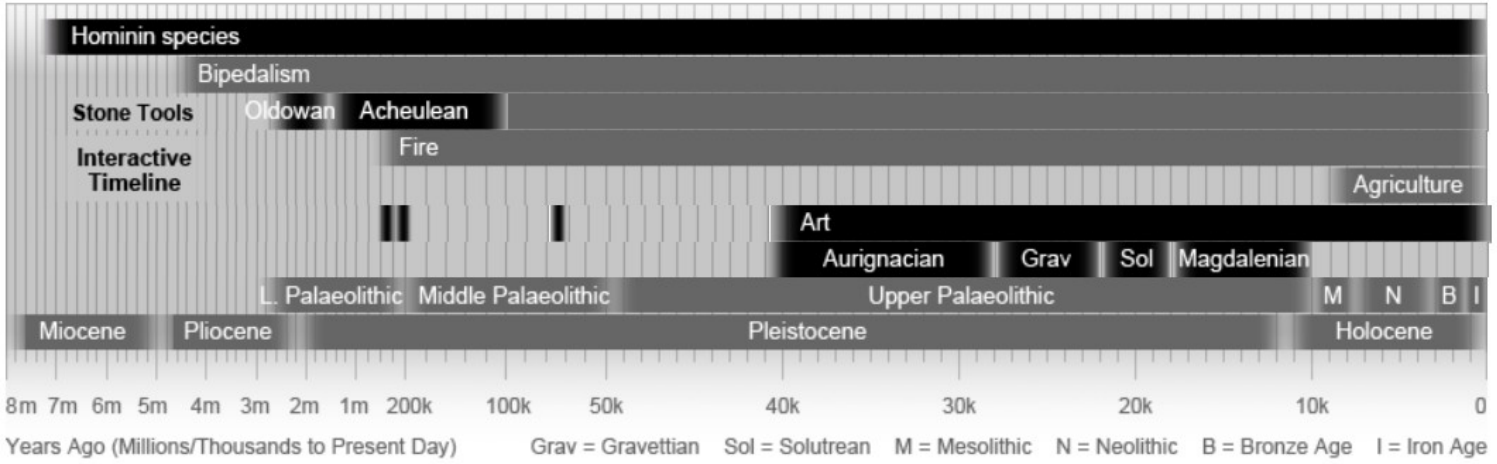
- Evolutionary transition between *H. heidelbergensis* and *H. sapiens* 300 000 – 100 000 years BP
- By 100 000 years BP: Early *Homo sapiens* – middle Palaeolithic
- Anatomically identical to modern humans
- Behaviourally not yet modern
- Hunter-gatherer societies
  - Used fire
  - Made sophisticated tools
  - Hunted and butchered animals
  - Made clothes
  - Had language
  - Made shelters





Cultural Evolution – *Homo sapiens* - upper Palaeolithic

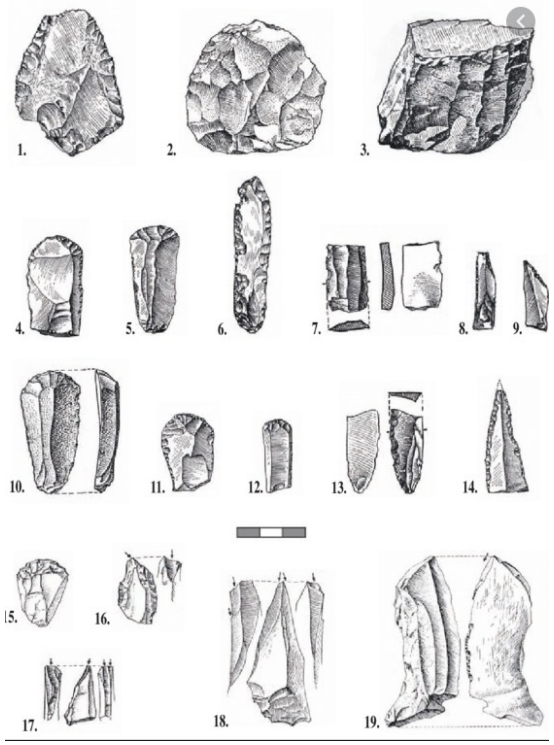
- 40 000 years – 10 000 years BP – “Cro-Magnon Man”
- Acceleration in technological development and migration
- Hunter-gatherer societies
  - Used fire
  - Made sophisticated tools
  - Hunted and butchered animals
  - Made clothes
  - Had language
  - Made shelters

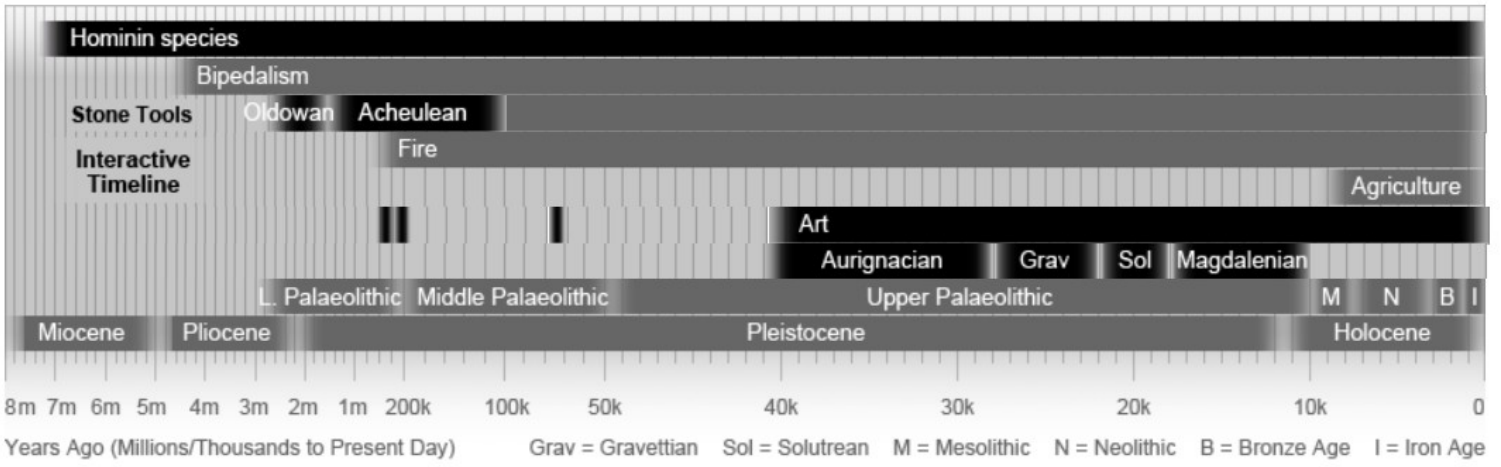


- Stone tools, hunting and butchering techniques, art and social structures increased in complexity.
- 3 main Cro Magnon cultures we look at in Human Biology:
  - Aurignacian (40 000 – 28 000 BP)
  - Solutrean (21 000 – 17 000 BP)
  - Magdelanian (17 000 – 12 000 BP)

Cultural Evolution – *Homo sapiens* - Aurignacian 40K – 28K BP

- Aurignacian tools:
  - Stone blade tools, made from flakes off large stone core, then finely shaped
  - Fine flaking
  - Variety or sizes, shapes and uses





- Aurignacian art

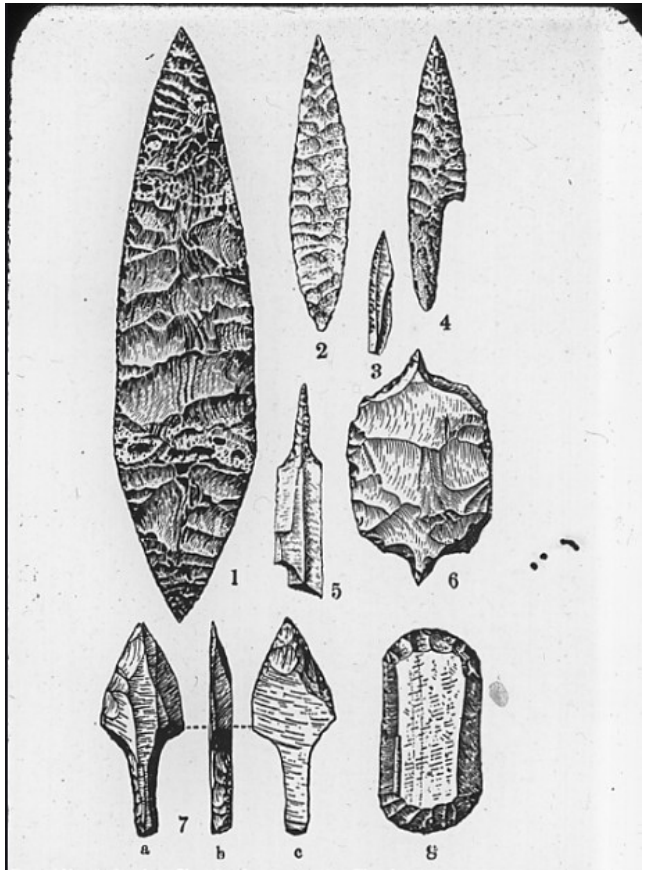


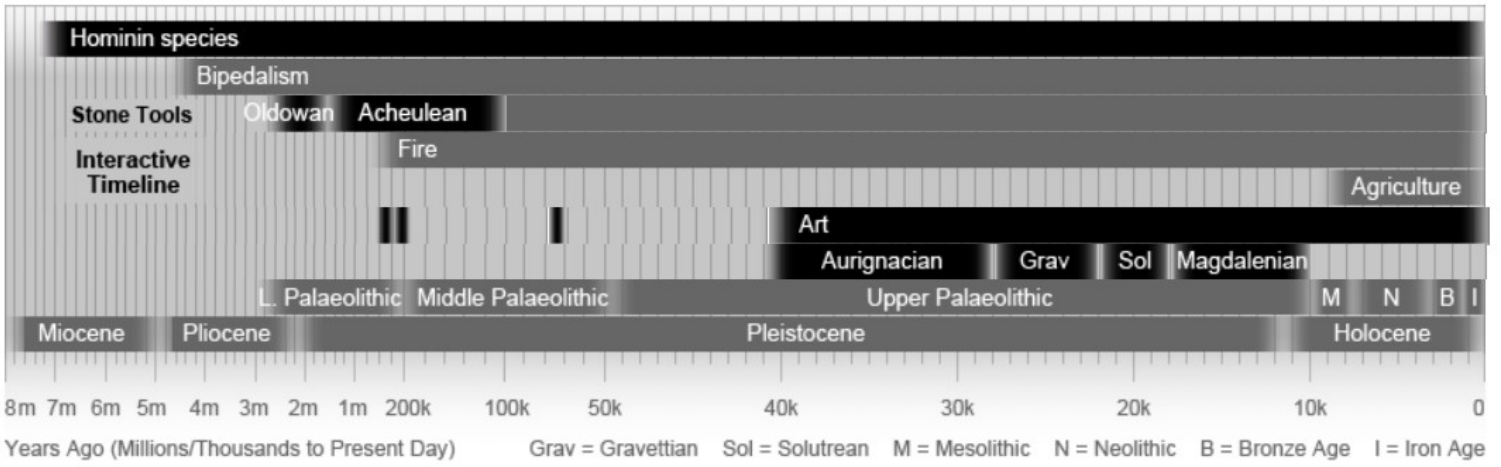
Cultural Evolution – *Homo sapiens* - Solutrean 21K – 17K BP

- Solutrean Tools
  - Very fine pressure flaking
  - Leaf-shaped blades
  - Symbolic blades

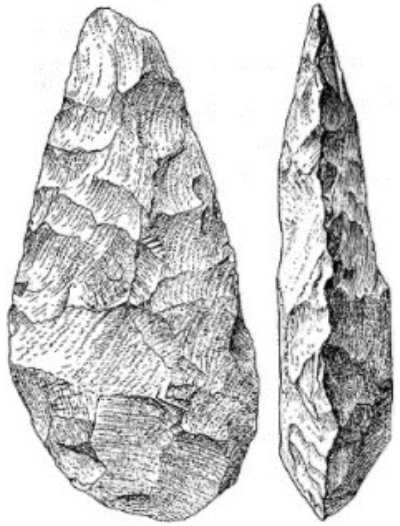


Figure 20.9 A Solutrean 'laurel-leaf' blade





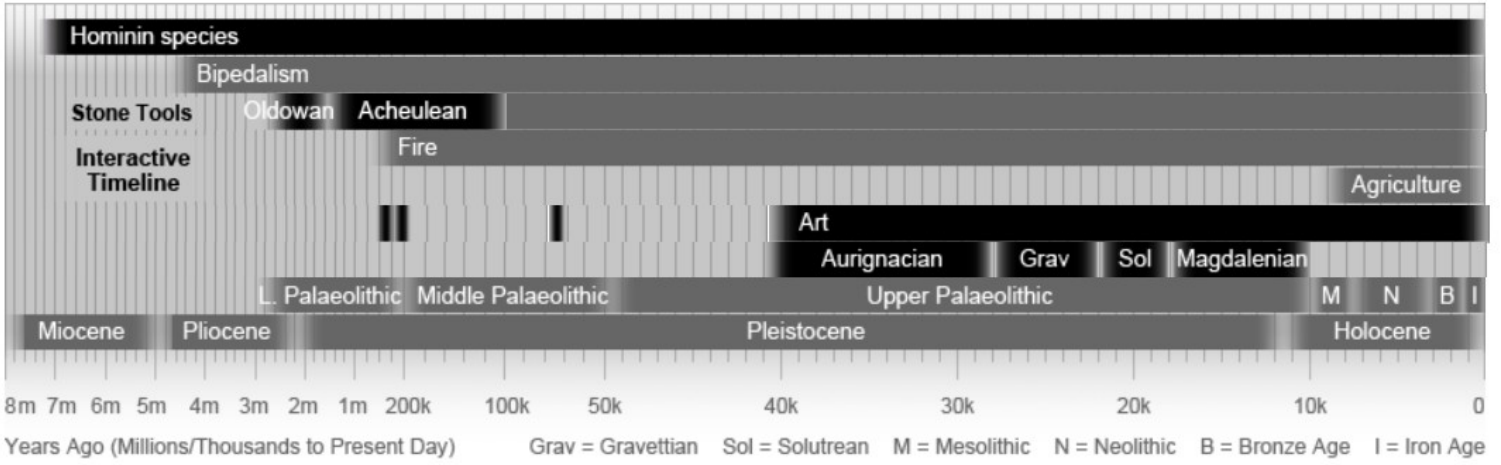
Some students get confused between the much older Acheulean hand axes used by H. erectus, and Solutrean leaf shaped blades. Solutrean blades have MUCH finer flaking, and are pointed at both ends.





Cultural Evolution – *Homo sapiens* - Magdelanian 17K – 12K BP

- Magdelanian Tools
  - Bone and antler tools predominated
    - Barbed spears
    - Spear throwers
    - needles
  - Tool for making other tools – a stone “burin” chisel used for carving bone and antler



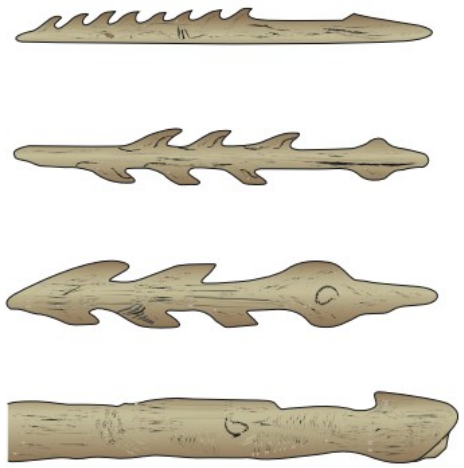


Figure 20.10 Magdalenian barbed points and spear thrower (bottom) made of bone or antler

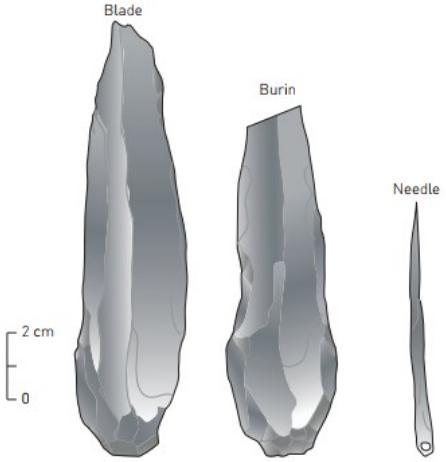


Figure 20.11 A burin was made from a basic blade. It was used to cut into antler and bone to make other tools, such as the needle.

- Art





Art works

Portable – carved figurines

Mural art on cave walls

Use of colour and shading



Figure 20.12 Cro-Magnon carving: **a** a Venus statuette carved from sandstone and **b** a horse carved from ivory



Philippe Psaila/Science Photo Library

Figure 20.13 Drawings representative of Cro-Magnon artwork

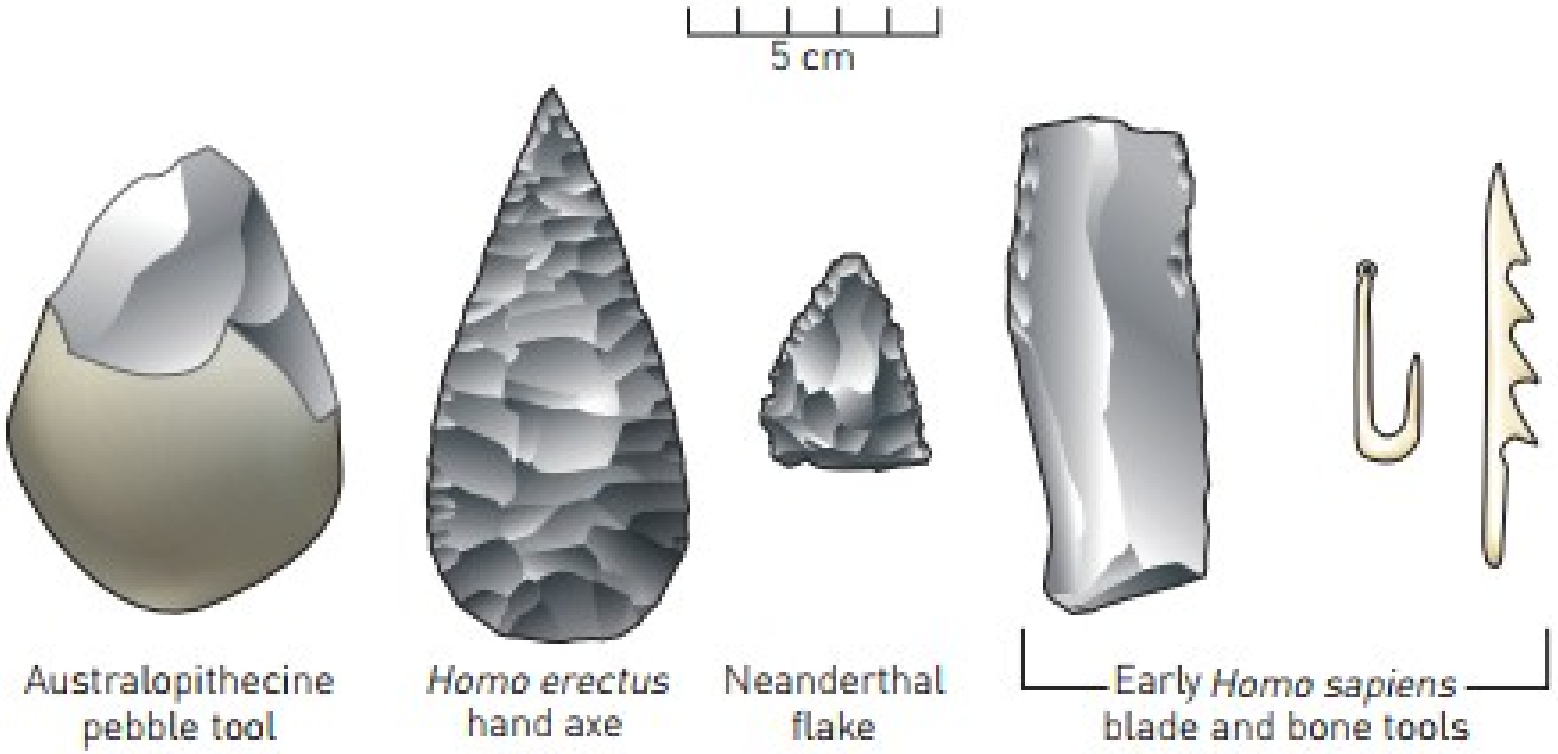


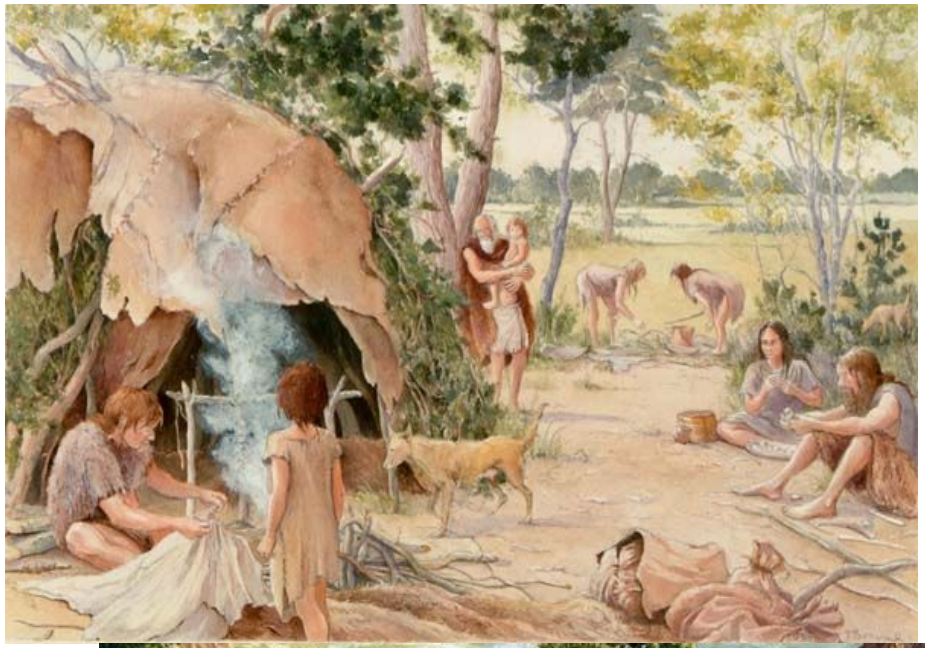
Figure 20.14 An example of cultural evolution – increasing sophistication of simple tools from australopithecines to early *Homo sapiens*

Note – as the tools become more modern, there is:

- Finer flaking
- Greater variety in shape and size
- Greater specialisation (eg blades, needles, spearheads etc)

Cultural Evolution – *Homo sapiens* - Mesolithic and Neolithic

- Big step in human history
- Change from nomadic to village lifestyle
- “Neolithic revolution”
- Domestication of plant and animal species
- Likely to have begun in South-West Asia, around Palestine
  - “Fertile Crescent”
  - 12 500 years ago
  - Gathered wild grasses, which hybridised
  - Fertile hybrid: “Emmer”
  - Later hybrid; large grains, good for making flour, but could no longer by scattered by wind; had to be sown – agriculture!
- Allowed easier gathering of resources and freed up time and energy for thinking, planning and innovation.





Cultural Evolution – *Homo sapiens* - Beginning of Agriculture

- Domestication of crops occurred independently at different sites around the world:
  - China – rice, millet, soybeans, yams
  - Central America – maize (corn), potato, sweet potato, bean, peanut, pumpkin, tomato
  - New World; more plants domesticated. Old World – more animals domesticated
- 11 000 years ago: animals domesticated
  - Sheep – south west asia
  - Pigs, cattle, goats, and other animals, over a longer period of time
- Farming supplemented by fishing and hunting
- Surplus grain could be stored
- More people could be supported on less land
- Settled life allowed development of new technologies



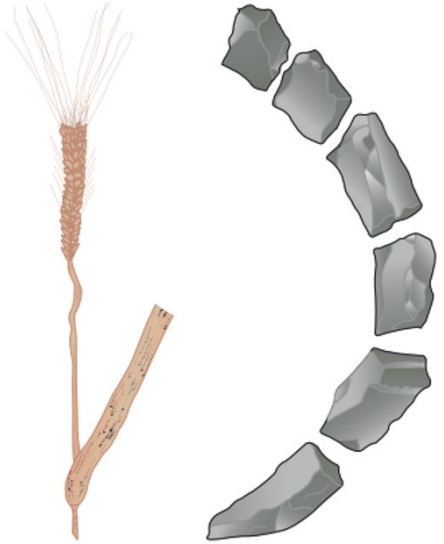


Figure 20.17 Wild wheat

Figure 20.18 Pieces of flint used as blades for a sickle to harvest wheat



Figure 20.19 The region of early settlement around the Persian Gulf and eastern Mediterranean, often referred to as the 'fertile crescent'

Cultural Evolution – *Homo sapiens* - The Modern Day

- We are able to manipulate our environment to a high level
  - Travel
  - Medical technology – life expectancy
  - Complex society
- Extremely fast acceleration in technology

- Important to understand human origins
  - Helps understand modern humans
  - Lifestyle diseases often because our cultural evolution has outstripped our biological evolution
- Still challenges to survival, some of our own making
  - Climate change an immediately pressing issue
  - May be able to evolve culturally so we are able to slow down or reverse the change we are making.




ARCTIC OCEAN
Greenland
Greenland Sea
Barents Sea
Kara Sea
Laptev Sea
East Siberian Sea
Chukchi Sea
Bering Sea
Beaufort Sea
Baffin Bay
Davis Strait
Norwegian Sea
Arctic Circle (66°33')
Alaska U.S.
Denali (6194 m)
Victoria Island
Great Bear Lake
Great Slave Lake
Hudson Bay
Labrador Sea
CANADA
North America
ROCKY MOUNTAINS
UNITED STATES
Lake Winnipeg
Lake Superior
Lake Huron
Lake Michigan
Lake Ontario
Lake Erie
Ottawa
Chicago
New York
Washington DC
Seattle
San Francisco
Death Valley
Los Angeles
Denver
Houston
Atlanta
Jacksonville
Mississippi
MEXICO
Gulf of Mexico
Mexico
Nassau
THE BAHAMAS
CUBA
Havana
DOMINICAN REPUBLIC
HAITI
Port-au-Prince
Santo Domingo
PUERTO RICO
JAMAICA
Kingston
BELIZE
GUATEMALA
HONDURAS
NICARAGUA
Caribbean Sea
COSTA RICA
Panama
PANAMA
VENEZUELA
Caracas
GUYANA
Georgetown
SURINAME
Paramaribo
FRENCH GUIANA
Cayenne
TRINIDAD AND TOBAGO
COLOMBIA
Bogotá
ECUADOR
Quito
Galapagos Islands
PERU
Lima
Marquesas Islands (Fr. Poly.)
Oceania
SOUTH PACIFIC OCEAN
Tropic of Capricorn (23°27')
Equator
Tropic of Cancer (23°27')
NORTH PACIFIC OCEAN
NORTH ATLANTIC OCEAN
Mid-atlantic Ridge
South America
Amazon River
BRAZIL
Brasília
BOLIVIA
PARAGUAY
Asunción
CHILE
Ojos del Salado (6,893 m)
Aconcagua (6,962 m)
Santiago
Buenos Aires
URUGUAY
Montevideo
ARGENTINA
ANDES
Argentine Basin
Tristan da Cunha (St. Helena)
Gough Island (St. Helena)
Falkland Islands
Stanley
Drake Passage
South Georgia and the South Sandwich Islands
Scotia Sea
Meteor Deep South Sandwich Trench (8,202m)
SOUTH ATLANTIC OCEAN
Mid-Atlantic Ridge
SOUTHERN OCEAN
Antarctic Circle (66°33')
Bellingshausen Sea
Amundsen Sea
Ross Sea
Ross Ice Shelf
Weddell Sea
Ronne Ice Shelf
Vinson Massif (4,892 m)
ANTARCTICA
Europe
Eurasia
North Asia
RUSSIA
Ural Mountains
SWEDEN
NORWAY
Oslo
Stockholm
FINLAND
Helsinki
Tallinn
ESTONIA
LATVIA
Riga
Moscow
UNITED KINGDOM
Dublin
IRELAND
London
North Sea
Copenhagen
Amsterdam
Berlin
GERMANY
POLAND
Warsaw
BELARUS
Minsk
Kyiv
UKRAINE
Paris
FRANCE
Munich
Bern
Vienna
ROMANIA
Bucharest
Black Sea
Caucasus Mts.
GEORGIA
Tbilisi
Yerevan
Baku
Caspian Sea
Aral Sea
KAZAKHSTAN
Central Asia
Astana
Lake Balkhash
Lake Baikal
Ulaanbaatar
MONGOLIA
Altai Mountains
Gobi Desert
Taklamakan Desert
Beijing
Hokkaido
Sea of Okhotsk
Sea of Japan
NORTH KOREA
Pyongyang
SOUTH KOREA
Seoul
JAPAN
Honshu
Tokyo
East Asia
CHINA
Wuhan
Yangtze
Shanghai
Taipei
Taiwan
Guangzhou
Hong Kong
Macau
Philippine Sea
Marianas Trench
Challenger Deep (10984 m)
FEDERATED STATES OF MICRONESIA
Palikir
MARSHALL ISLANDS
Majuro
Tarawa
KIRIBATI
Oceania
Manila
PHILIPPINES
Saipan
Hagåtña
Palau
South China Sea
Philippine Sea
Celebes Sea
Andaman Sea
Gulf of Thailand
Bay of Bengal
Arabian Sea
Indochina
THAILAND
CAMBODIA
Phnom Penh
VIETNAM
Hanoi
LAOS
Vientiane
Yangon
MYANMAR (BURMA)
Naypyidaw
BANGLADESH
Dhaka
Kolkata
Kathmandu
Mount Everest (8,848 m)
Himalaya Mts.
Tibetan Plateau
Lake Namtso
Yarlung Tsangpo
Brahmaputra
Ganges
Varanasi
Delhi
Kashmir
Amritsar
Jaipur
Agra
Mumbai
Hyderabad
Chennai
Bangalore
Kochi
SRI LANKA
Colombo
MALDIVES
Male
Andaman Islands
Nicobar Islands
Southeast Asia
MALAYSIA
Kuala Lumpur
Singapore
Borneo
Sumatra
INDONESIA
Jakarta
Java
Java Trench
Timor
Banda Sea
Ceram Sea
Halmahera
Sulawesi
Celebes Sea
Molucca Sea
Flores Sea
Lombok
Bali
Komodo
Sumbawa
Ombai
Timor Sea
Arafura Sea
PAPUA NEW GUINEA
Port Moresby
New Guinea
Solomon Islands
Honiara
Coral Sea
Vanuatu
Port Vila
Fiji
Suva
NEW CALEDONIA
Nouméa
TONGA
SAMOA
Apia
TUVALU
Funafuti
Pacific Ocean
Australasia
AUSTRALIA
Deserts of Australia
Alice Springs
Darwin
Perth
Great Australian Bight
Brisbane
Sydney
Canberra
Melbourne
Tasman Sea
NEW ZEALAND
Wellington
Auckland
Christchurch
Hobart
Tasmania
Yaren District
Nauru
Indian Ocean
INDIAN OCEAN
Arabian Sea
Bay of Bengal
Andaman Sea
Central Indian Ridge
Carlsberg Ridge
Chagos-Laccadive Ridge
Ninetyeast Ridge
Southwest Indian Ridge
Kerguelen Plateau
Mascarene Basin
Crozet Basin
Madagascar Basin
Wharton Basin
Java Trench
Sunda Trench
Tropic of Capricorn (23°27')
Equator
Tropic of Cancer (23°27')
Antarctic Circle (66°33')
Middle East
SAHARA
ALGERIA
LIBYA
EGYPT
Cairo
Nile River
Aswan High Dam
Lake Nasser
Western Sahara
MAURITANIA
Nouakchott
MALI
NIGER
CHAD
SUDAN
Khartoum
Niamey
Bamako
Dakar
Praia
CAPE VERDE
GUINEA-BISSAU
GUINEA
Conakry
SIERRA LEONE
Freetown
LIBERIA
Monrovia
CÔTE D'IVOIRE
Yamoussoukro
Abidjan
GHANA
Accra
TOGO
Lomé
BENIN
Porto-Novo
Cotonou
NIGERIA
Abuja
Lagos
CAMEROON
Yaoundé
Douala
EQUATORIAL GUINEA
Malabo
GABON
Libreville
CONGO
Brazzaville
CAR
Bangui
SOUTH SUDAN
Juba
ETHIOPIA
Addis Ababa
SOMALIA
Mogadishu
Africa
DEMOCRATIC REPUBLIC OF THE CONGO
Kinshasa
Lake Tanganyika
Lake Victoria
Lake Turkana
Lake Albert
Lake Kivu
UGANDA
Kampala
KENYA
Nairobi
Kilimanjaro (5,895 m)
RWANDA
BURUNDI
Gitega
TANZANIA
Dar es Salaam
Dodoma
Lake Malawi
Lake Nyasa
ANGOLA
Luanda
ZAMBIA
Lusaka
Lake Kariba
ZIMBABWE
Harare
MOZAMBIQUE
Maputo
Beira
Madagascar
Antananarivo
COMOROS
Moroni
MAURITIUS
SEYCHELLES
Victoria
NAMIBIA
Kalahari Desert
Windhoek
BOTSWANA
Gaborone
Pretoria
SOUTH AFRICA
Johannesburg
ESWATINI
LESOTHO
Maseru
Orange River
Namib Desert
Walvis Bay
Cape Town
Cape of Good Hope
Cape Agulhas
Port Elizabeth
Durban
Richards Bay
Kruger National Park
Limpopo River
Zambezi River
Victoria Falls
Okavango Delta
Etosha Pan
Caprivi Strip
Cunene River
Kunene River
Namibe
Lobito
Benguela
Cabinda
Pointe-Noire
Brazzaville
Kinshasa
Matadi
Bangui
Juba
Khartoum
Omdurman
Port Sudan
Red Sea
Gulf of Aden
Socotra
Mogadishu
Kismayo
Nairobi
Mombasa
Dar es Salaam
Zanzibar
Lake Victoria
Lake Tanganyika
Lake Malawi
Lake Kariba
Lake Turkana
Lake Albert
Lake Kivu
Kilimanjaro (5,895 m)
Mount Kenya
Serengeti
Ngorongoro
Rift Valley
Great Rift Valley
Ethiopian Highlands
Lake Tana
Blue Nile
Sahara
Nile River
Aswan High Dam
Lake Nasser
Suez Canal
Sinai Peninsula
Gulf of Suez
Gulf of Aqaba
Dead Sea
Jordan River
Jerusalem
Amman
Petra
Wadi Rum
Aqaba